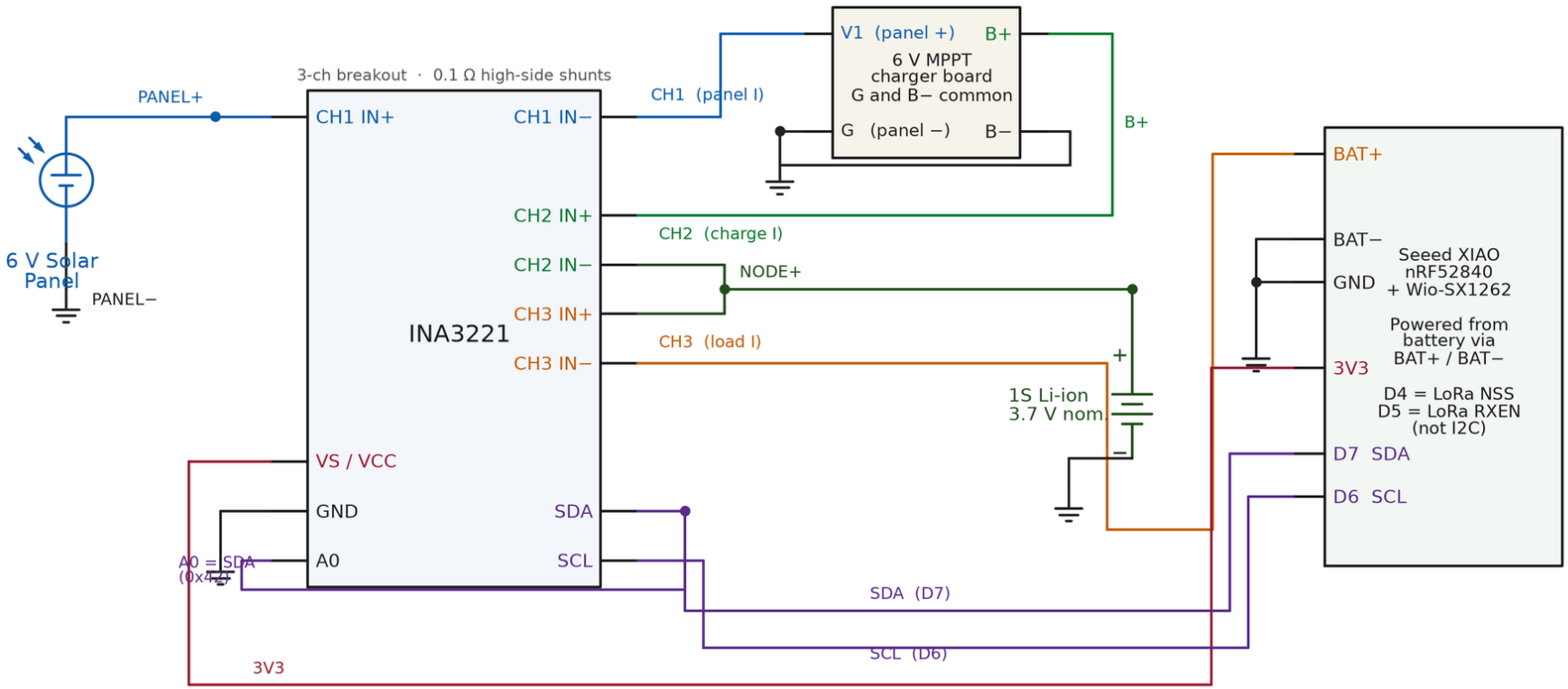
Solar Monitor Add-on — MeshCore Repeater Node

6 V panel · INA3221 3-ch high-side · 6 V MPPT · 1S Li-ion · XIAO nRF52840 + Wio-SX1262



Channel map

CH1 = panel V and A
CH2 = MPPT output V and A (charge current)
CH3 = node load V and A
Compare CH1 with CH2 to tell whether the panel or the MPPT failed.
Max about 1.6 A per channel with the 0.1 Ω shunts on the breakout.
All grounds are common.

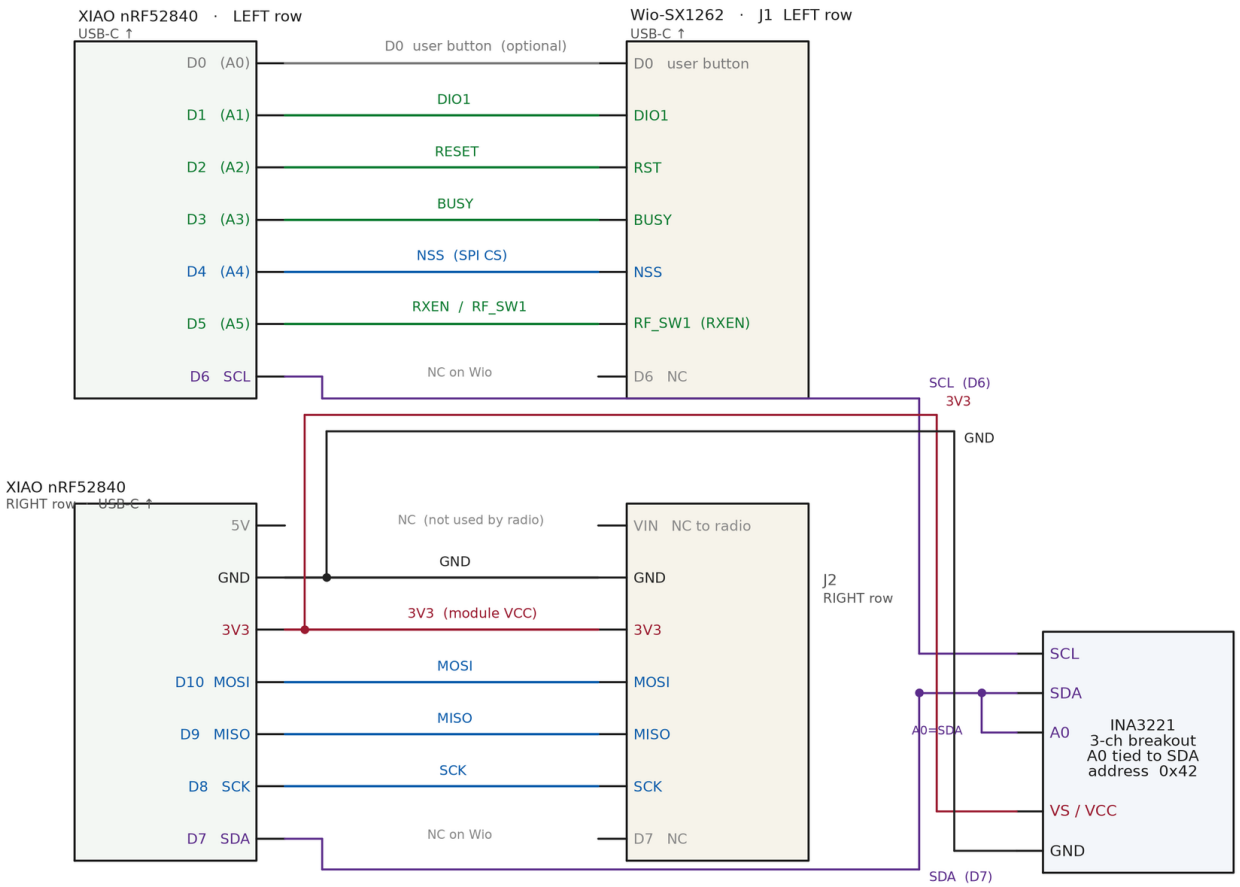
I2C and address (MeshCore firmware)

SDA = XIAO D7, SCL = XIAO D6. Default XIAO I2C pins D4 / D5 are taken by the Wio-SX1262 (NSS / RXEN). Pins from variants/xiao_nrf52/platformio.ini: PIN_WIRE_SDA=D7, PIN_WIRE_SCL=D6, P_LORA_NSS=D4, SX126X_RXEN=D5.
A0 is tied to SDA (address 0x42), the stock MeshCore TELEM_INA3221_ADDRESS. Breakouts often ship with A0 at GND (0x40), which is the INA219 default; move A0 to SDA so the INA3221 is found at 0x42.

XIAO nRF52840 ↔ Wio-SX1262 point-to-point wiring

Seeed XIAO-format Wio-SX1262 (through-hole headers, not the ESP32-S3 B2B board) · MeshCore xiao_nrf52

Header order is the physical XIAO footprint, USB-C at the top. Wio J1/J2 sit 1:1 on those pins when stacked.



What MeshCore actually uses

Radio: D1 DIO1, D2 RESET, D3 BUSY, D4 NSS, D5 RXEN, D8 SCK, D9 MISO, D10 MOSI, plus 3V3 and GND.
TXEN is not a host pin (SX126X_TXEN=NC). DIO2 inside the module drives the TX RF switch (DIO2_AS_RF_SWITCH=1).
D5 is RXEN (RF_SW1) and has a 10 kΩ pull-up on the Wio.
DIO3 is the module TCXO supply (1.8 V) and is not brought out.
The Wio carrier has 22 Ω series resistors on SPI and NSS.
I2C: SDA=D7, SCL=D6. INA3221 A0 is tied to SDA (0x42), the stock MeshCore TELEM_INA3221_ADDRESS.

Free pins and wiring notes

Free on the header: D6 and D7 (I2C to the INA3221), D0 (optional Wio user button; radio does not need it), and 5V/VIN (header pin only; the module runs from 3V3).
BAT+ / BAT− are underside pads on the XIAO, not these headers.
Wio pin numbering matches the XIAO physically, 1:1, when stacked (J1 = left row, J2 = right row, pin 1 at USB-C).
This is the nRF52840 through-hole Wio, not the ESP32-S3 B2B board — those pinouts are different. Keep SPI wires (SCK, MOSI, MISO, NSS) short and of similar length.

INA3221 board
MusRock INA3221 3-pack (Amazon ASIN B0FTF9LC1T). Marked 100 mΩ (R100) shunts, which match MeshCore's default 0.1 Ω. Verify the shunts actually read R100 before stuffing the board.

I2C address
The breakout ships at 0x40. Move A0 to SDA for 0x42 — that is the address stock MeshCore probes for the INA3221 (TELEM_INA3221_ADDRESS). 0x40 is the INA219 default and can be misdetected as one.

I2C pins
D7 = SDA, D6 = SCL. D4 and D5 are the radio's NSS and RXEN, not I2C. Source: MeshCore variants/xiao_nrf52/platformio.ini (PIN_WIRE_SDA=D7, PIN_WIRE_SCL=D6, P_LORA_NSS=D4, SX126X_RXEN=D5).

Channel map
CH1 = panel V and A. CH2 = MPPT output V and A (charge). CH3 = node load V and A. CH1 power with CH2 near zero means the MPPT is the problem; low CH1 in sun means the panel.

Shunt rating
Max about 1.6 A per channel with the 0.1 Ω shunts.

MPPT pads
V1 = panel +. G = panel −. B+ / B− = battery. G and B− are assumed common — confirm with a continuity check on the actual board.

Radio wiring
Used: D1 (DIO1), D2 (RESET), D3 (BUSY), D4 (NSS), D5 (RXEN), D8 (SCK), D9 (MISO), D10 (MOSI), 3V3, GND. Free: D0 (Wio user button only), D6, D7, 5V. TX switch is DIO2 inside the module; there is no TXEN pin.

Firmware
Stock MeshCore — and the v1.18.0-jp.wdt build based on it — already includes INA3221 support. Bench-test with a telemetry pull first; no firmware change is expected.

Telemetry
CayenneLPP power reports in whole watts, so small loads can show 0 W even when V and A are correct.